Swarnim Barapatre

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Professional Summary

Software engineer with two years of experience building backend services, full-stack applications, and LLM-based agents for industrial and autonomous-systems software. Works mainly in TypeScript/JavaScript, Python, and Go, and uses Rust for desktop and systems projects.

Professional Experience

Coditation Systems (Plant360.ai) Pune, Maharashtra
AI Full-Stack Software Engineer Feb, 2026 — Present

- Built an AI agent that generates safe-shutdown and maintenance plans for industrial equipment from Plant360 graphs. Validation gates and audit trails let engineers review and trust every plan.
- Built and deployed production backend API around the agent using FastAPI, PostgreSQL, background workers, streaming updates over SSE, and audit logs.
- Built conversion services that extract proprietary P&ID drawings into the Plant360 graph format (reverse-engineering SmartPlant's API) and export them to DEXPI XML and AutoCAD DWG via async REST APIs.
- Built a library in React and TypeScript that renders Plant360 graphs as interactive P&ID diagrams, now used across the organization's products.
- Built a cross-platform desktop app in Rust and Tauri that validates and renders DEXPI engineering-diagram files against the schema and the official spec.

AccurateIC Pune, Maharashtra
Full-Stack Software Engineer, AI Department Aug, 2024 — Feb, 2026

- Architected and developed ISACA, a full-stack monorepo for Indian Navy autonomous ship-control workflows, integrating ROS middleware through custom pub-sub transport and real-time multi-sensor fusion across radar, GPS, cameras, eMRU, and odometry streams.
- Built radar overlay, AIS tracking, dynamic route planning, waypoint management, voyage tracking, and shore-to-ship synchronization using bidirectional WebSocket architecture.
- Developed real-time shell command execution and monitoring flows with UI feedback, and set up Jenkins CI/CD with automated database migrations, seeding, and systemd orchestration.
- Led full-stack development for the BlueGulf x AccurateIC collaboration, adapting autonomous navigation systems for high-speed vessels and showcasing the platform at DriftX Abu Dhabi Autonomous Expo at Yas Marina.
- Architected and developed NeuroGen, an IoT-enabled genset monitoring platform (Go/Chi backend, React frontend) with predictive maintenance, multi-variate anomaly detection, life expectancy simulations, and operational data visualizations.
- Drove architecture, code reviews, junior developer mentoring, and performance profiling, achieving 40% CPU reduction in critical application flows.

MasterSoft ERP Solutions Nagpur, Maharashtra
Software Engineering Intern, R&D Department Aug, 2023 — June, 2024

- Developed Panopticon, a multi-tenant employee productivity measurement system serving 700+ concurrent users, eliminating DeskTime dependency and saving ₹53 lakh annually.
- Implemented Panopticon features including atomic app updates, live configuration sync, encrypted activity logs, and accurate productivity metrics for remote talent evaluation.
- Built an internal MS-SQL suite for database developers, reducing SQL task flows by 85% through stored procedure diffs, alter-script generation, and memory leak detection tooling.
- Built comprehensive internal developer toolchains and automation frameworks, reducing development cycle times by up to 14 days and improving team productivity across multiple projects.
- Maintained critical Linux server infrastructure, implemented proactive monitoring solutions, and resolved production incidents to ensure system uptime and reliability.

Education

BITS Pilani Vasco da Gama, Goa
B.E. Chemical Engineering + M.Sc. Biological Sciences May, 2019 — July, 2024

Skills

- **Languages**: TypeScript, JavaScript, Python, Go, Rust, SQL, Bash, PowerShell
- **Frontend**: React, Zustand, Redux, TanStack Query, HTML, CSS, TailwindCSS, Tauri
- **Backend**: AdonisJS MVC, Express.js, Node.js, OAuth, RabbitMQ, Lucid ORM, Chi, SQLAlchemy, Celery, Flask, FastAPI, Django, WebSockets, Server-Sent Events
- **AI/ML**: AI Agents, Tool Calling, Agentic Loops, Anthropic API, OpenAI API, Gemini API, Self-Hosted LLMs
- **Databases**: PostgreSQL, SQLite, Microsoft SQL Server, MongoDB
- **Cloud/DevOps**: AWS EC2, Docker, Nginx, GitLab, GitHub, GitHub Actions, Jenkins, Linux
- **Other**: ROS, Emacs Lisp

Personal Projects

Metamorph — Workout Tracking App Oct, 2026 — Present

- Built and deployed a full-stack workout tracker as a web and Android app, with Google sign-in, account deletion, and per-workout time zones. Live at metamorph.smarniw.com.
- Rust API (Axum, SQLx) on PostgreSQL with migrations and integration tests; CI on GitHub Actions with gated deploys and rollback.
- Tech Stack: *Rust, Axum, PostgreSQL, React, TypeScript, Tauri*

Fero — Continuation of the RQuickShare Project May, 2026 — June, 2026

- Forked and continued RQuickShare, an open-source Quick Share client for Linux and macOS that had been unmaintained for two years.
- Migrated the frontend from Vue to React + Tailwind, and added clipboard text and image sharing over the Quick Share protocol.
- Modernized the CI and release pipeline on GitHub Actions.
- Tech Stack: *Rust, Tauri, React, TypeScript, GitHub Actions*

Spotagooey — Desktop Spotify Client Aug, 2026

- Desktop Spotify client with native playback through librespot, Spotify Connect control, synced lyrics, and a command palette.
- Tech Stack: *Rust, Tauri, React, TypeScript*

Turepo — Package for Emacs Dec, 2025 — Present

- Built a developer productivity package for the GNU Emacs text editor.
- Passed MELPA's code & quality standards. Package merged in MELPA.
- Tech Stack: *Emacs Lisp*

Kafkaesqueue — Custom Queueing System Oct, 2023 — June, 2024

- Lightweight queue management system with FIFO guarantees.
- Opt-in parallel task processing using threads.
- Built-in persistence via SQLite or in-memory storage.
- Tech Stack: *Python, FastAPI, Pydantic*

Extracurriculars

Swimming: National Swimmer. Multiple-time State Gold Medalist.